HOSPITAL DATA ANALYTICS PROJECT

Power BI, DAX, Python, SQL and Data Visualization

Project Report

Prepared by: Clement Jonathan

Data Analytics Training Project

Torbita Computer Training School, Abuja

This report documents the data preparation, modelling, analysis and visualization of hospital datasets using Microsoft Power BI and related analytics tools.

1. Executive Summary

This project demonstrates the use of data analytics techniques to transform hospital-related data into useful management information. The analysis covers data cleaning and transformation in Power Query, data modelling in Power BI, DAX-based calculations and interactive dashboard development. The main analytical areas include patient volume, treatment cost, revenue, patient type, financial classification, waiting time, consultation duration and overall visit duration.

2. Project Objectives

The objectives of the project were to: (1) obtain and prepare real-world datasets; (2) import and clean the datasets using Power Query; (3) model the data for analysis; (4) develop at least 15 DAX calculations; and (5) create clear and interactive dashboards.

3. Data Sources and Preparation

The project used hospital datasets obtained from publicly available datasets. The datasets were inspected for data types, missing values, inconsistent entries, duplicate or unsuitable fields, and numerical fields stored in inappropriate formats. Power Query was used to transform the data before loading it into Power BI Desktop.

4. Dataset 1: Hospital Data Analysis

The first dataset was used to examine patient-level hospital information. Key analytical fields included patient identification, demographic information, clinical condition, admission information, length of stay, treatment cost, outcome, readmission and satisfaction-related information.

Key analytical measures developed

Measure	Purpose
Total Patients	Counts the patients represented in the dataset.
Average Age	Calculates the mean patient age.
Total Cost	Calculates total treatment cost.
Average Cost	Calculates average treatment cost.
Average Stay	Measures average length of hospital stay.
Maximum / Minimum Cost	Identifies the highest and lowest treatment costs.
Readmitted Patients	Counts patients recorded as readmitted.
Readmission Rate	Measures the proportion of patients readmitted.
Average Satisfaction	Measures average patient satisfaction.
Recovered Patients	Counts patients with a recovered outcome.
Recovery Rate	Measures the proportion of recovered patients.
Total Stay	Calculates the combined length of stay.
Cost per Day	Estimates treatment cost per day of stay.
High Cost Patients	Identifies patients above a defined cost threshold.

5. Dataset 2: Hospital Operations and Revenue Analysis

The second dataset provided additional operational and financial information. Important fields included Lab Cost, Consultation Revenue, Patient Type, Financial Class, Date, Entry time, Post-Consultation Time and Completion Time. The time fields contain the same date with different times, allowing the analysis to examine the movement of a patient through the service process.

Operational Time Analysis

Calculated fields were developed to estimate waiting time from Entry time to Post-Consultation Time, consultation duration from Post-Consultation Time to Completion Time, and total visit time from Entry time to Completion Time. Average waiting, consultation and visit times can then be used as dashboard KPIs.

Financial and Patient Analysis

The dataset supports analysis of laboratory cost and consultation revenue, as well as comparison between INPATIENT and OUTPATIENT records. Financial Class can be used to compare patient volumes and revenue across financial categories.

6. Dashboard Design

The dashboards were intentionally designed to be simple and readable because the project was developed on a system with limited screen space. The main dashboard uses KPI cards and a small number of charts rather than overcrowding the report page.

Visual	Analysis
Total Patients card	Overall patient volume.
Total Revenue card	Overall revenue generated from selected revenue fields.
Average Waiting Time card	Average time before post-consultation processing.
Average Visit Time card	Average overall visit duration.
Revenue by Financial Class	Compares revenue across financial categories.
Revenue Trend	Shows how revenue changes over time.
Patient Type Distribution	Compares INPATIENT and OUTPATIENT records.
Patients by Financial Class	Shows patient distribution across financial classes.

7. Data Modelling

The data was structured in Power BI so that numerical fields could be aggregated and categorical fields could be used for grouping and filtering. Date fields were prepared for trend analysis, while patient and financial categories were retained as dimensions for dashboard analysis. Where appropriate, calculated measures were preferred for summary KPIs, while row-level duration calculations were created as calculated columns.

8. Key Findings and Insights

The project shows that hospital data can be transformed into several useful performance indicators. Patient volume can be monitored through total and patient-type measures. Financial analysis can identify differences in revenue across financial classes and revenue categories. Operational time measures can highlight how long patients spend moving through the service process. Trend analysis provides a way to observe changes in activity over time, while outcome and readmission measures can support quality-of-care discussions.

9. Recommendations

The hospital can use the dashboard to monitor patient activity and financial performance regularly. Management should pay attention to unusually long waiting or visit times, investigate financial classes or services with unusual revenue patterns, and use readmission and outcome indicators as supporting quality measures. Data-entry standards should also be maintained so that dates, times, costs and patient categories remain consistent. Regular review of the dashboard would make the analysis more useful for operational decision-making.

10. Challenges Encountered

During the project, system and software limitations affected some of the practical work. The work was completed using available workarounds while continuing with the Power BI, DAX and data-analysis components. Working with imperfect real-world data also required attention to data types, missing or problematic values and aggregation logic.

11. Conclusion

This project demonstrates an end-to-end data analytics workflow using hospital data. It combines data cleaning, modelling, DAX calculations and dashboard design. The resulting analysis provides a practical example of how healthcare data can be converted into understandable information for monitoring patient activity, financial performance and operational efficiency.

Project Portfolio Note

This report is prepared as a portfolio document to accompany the Power BI project files. The GitHub repository can contain this PDF together with the Power BI report, datasets where redistribution is permitted, Python scripts and SQL files.